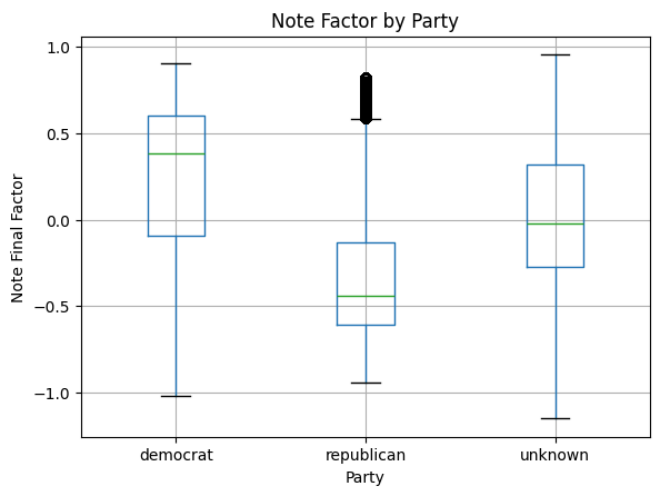
Partisanship

1. Does partisanship as measured through note factors align with partisanship as measured through the external data I sent you (from the trenault GitHub)?

To measure alignment, I encoded party as a numeric variable (Republican = -1, Democrat = 1, Unknown = NaN) and calculated the correlation with noteFinalFactor.

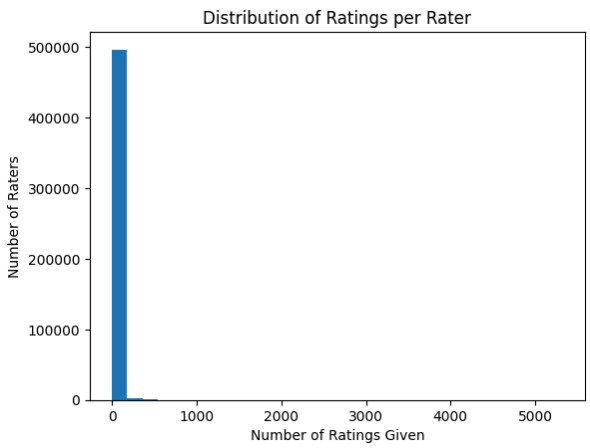
	party_num	noteFinalFactor
party_num	1.000000	0.572499
noteFinalFactor	0.572499	1.000000

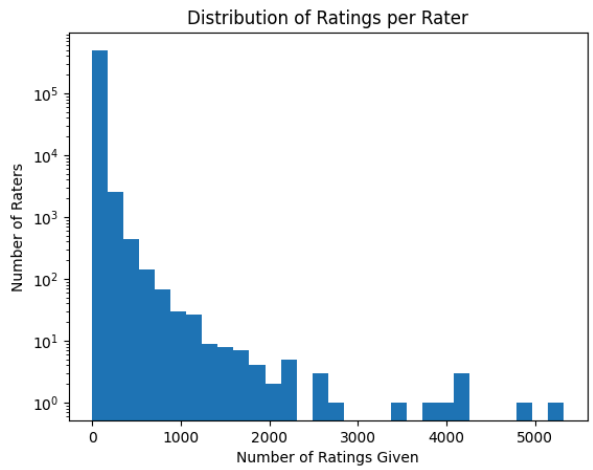


Correlation = 0.57, indicating a moderate-strong positive relationship. As the party shifts from Republican to Democrat, the note factor tends to increase from negative to positive. This indicates that notes about Republican tweets tend to have negative factors, while notes about Democrat tweets tend to have positive factors. This is also portrayed in the boxplot where the Democrat's box sits above zero, the Republican box sits below zero, and the unknown box sits right around zero.

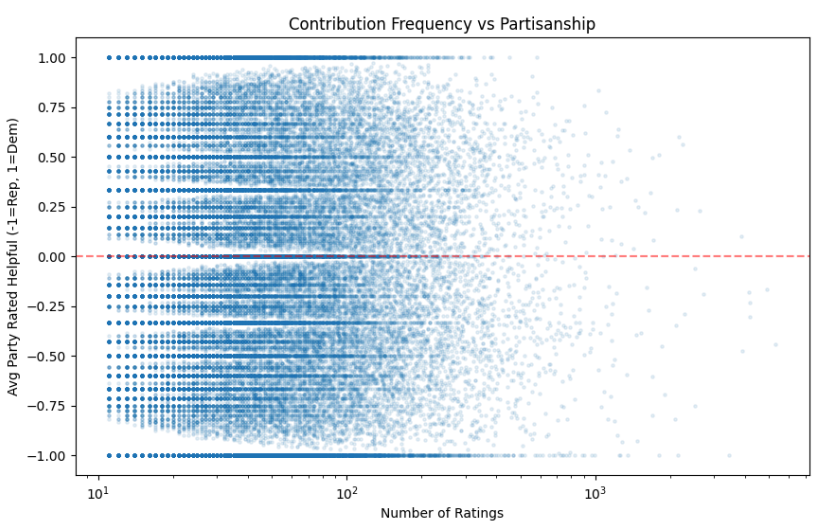
2. What is the relationship between how frequently a user contributes and their partisanship?

To begin, I examined the distribution of ratings per rater. There were 499,818 unique raters in May 2024. The distribution is highly skewed with 30% of raters, 149,727 users, submitted only a single rating, and the median was 3 ratings per rater. 75% of raters submitted 11 or fewer ratings. This indicates that for most users, partisanship signals calculated from their ratings would not be meaningful. Therefore, I filtered to "active raters" with more than 10 ratings to ensure reliable signal.





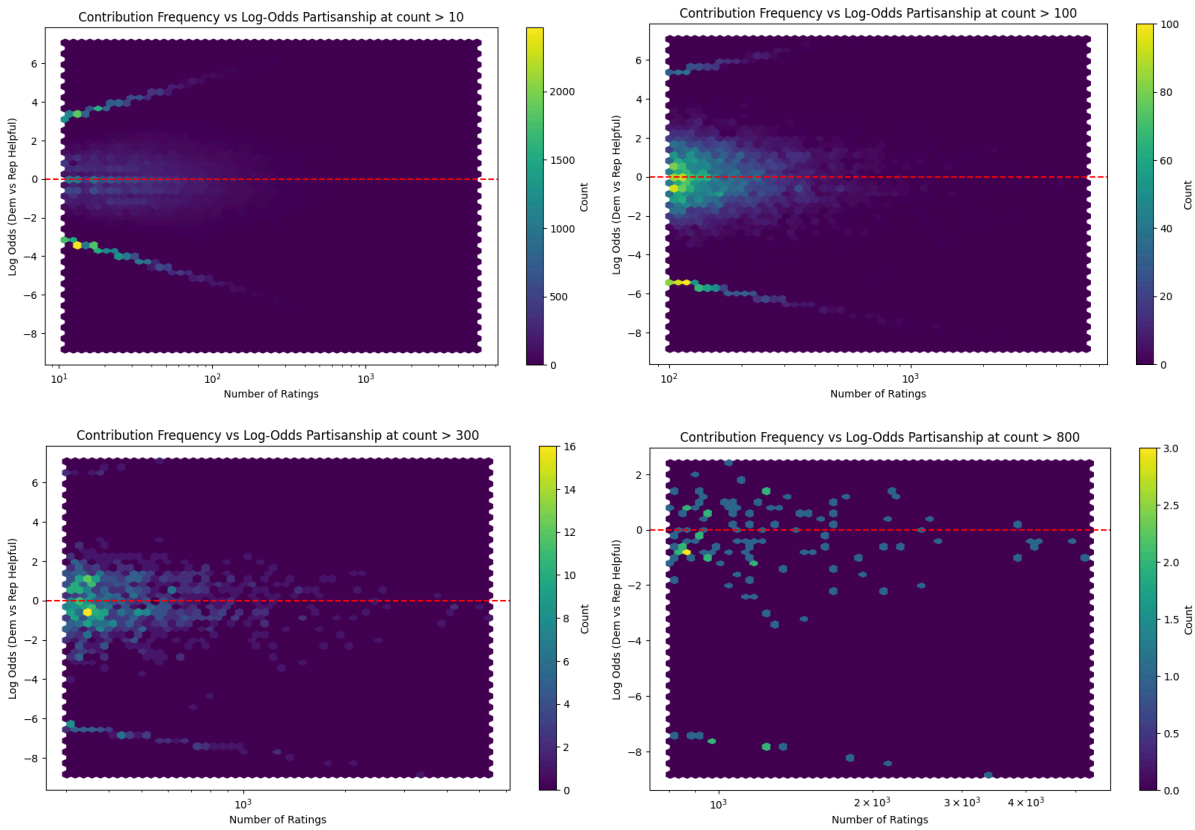
To explore the relationship between contribution frequency and partisanship, I first plotted a scatter plot of rating count vs average party rated helpful. However, the scatter plot was too dense to interpret clearly.



The clear horizontal lines visible at low rating counts are a mathematical artifact of how the party is encoded as -1 and +1. Because each rating changes the sum by exactly 2 (either +1 or -1), the possible average values are constrained to discrete steps. For example with 13 ratings, the numerator can only be odd numbers (-13, -11, -9... +13), so the only possible averages are -13/13, -11/13, -9/13, etc. Since many users

share the same low rating count, thousands of dots pile up at the same discrete y-values, creating the visible horizontal bands. As rating count increases, the steps get smaller and the bands fade into a continuous cloud.

To better visualize the density, I switched to a hexbin plot which shows how many users fall in each region using color. I also converted the partisanship measure to log odds where instead of using the raw percentage of Republican vs Democrat helpful ratings, I computed the ratio of Democrat to Republican helpful ratings and took the log. This makes the scale symmetric around 0 (0 = balanced, positive = Democrat leaning, negative = Republican leaning) and better captures extreme partisan users who would otherwise be squished at the edges of a -1 to 1 scale.



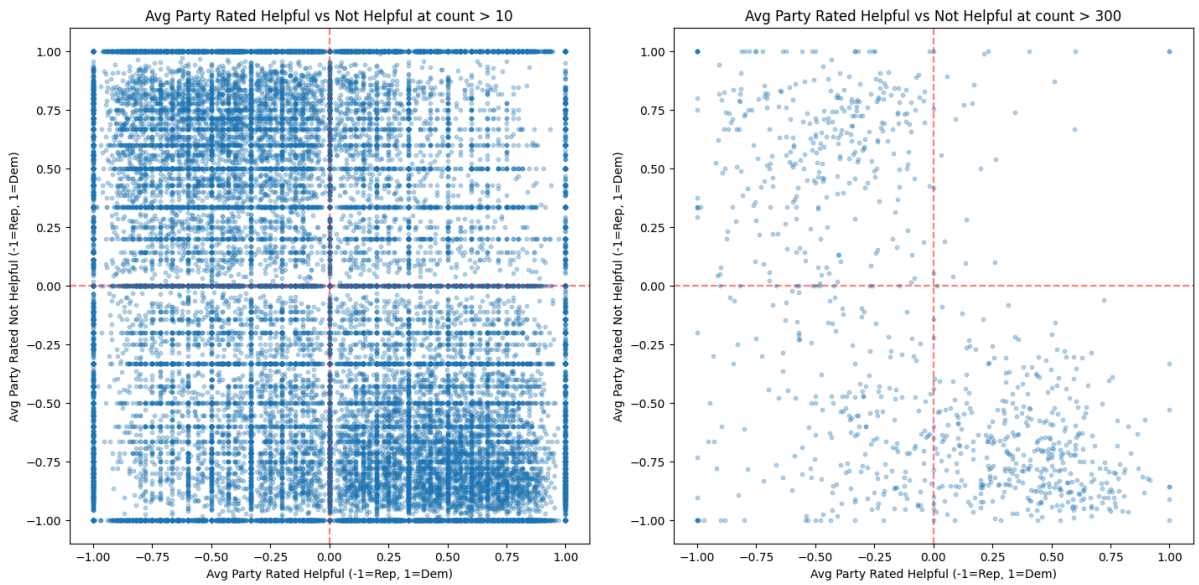
The four charts describe contribution frequency compared to log-odds of partisanship at different rating thresholds, showing how the distribution shifts depending on how active a user is.

The largest single group in the entire dataset is one-time raters, with nearly 18,000 people making just one rating, and that rating leaned slightly Republican-helpful. However, this is likely a reflection of the dataset having more Republican contributors overall rather than a behavioral pattern. As we look at more active raters, 9,211 people

with 100+ ratings, then 1,098 with 300+, and finally just 127 people with 800+ ratings, the distribution becomes more and more balanced and centered near zero. Overall, while the dataset has a slight Republican-leaning bias, the more engaged a contributor is, the more evenly they tend to rate both parties.

For example, if you rate 500 notes, you're probably encountering notes about both parties just by chance. It is almost impossible to only see Republican or Democrat tweets when you're rating that many notes.

3. Do users systematically rate one side helpful and the other not helpful?



Yes, users systematically rate one side helpful and the other not helpful. Active raters cluster strongly in the top-left (pro-Republican) and bottom-right (pro-Democrat) quadrants, indicating that partisan bias in ratings is a real and common pattern among the contributors.

Question 2 indicates that frequency and partisanship are negatively correlated where more active raters have more balanced on average, but Question3 adds on to it by saying “more balanced” does not mean unbiased. Systematic partisan raters exist even among highly active users.

At number of ratings > 10	At number of ratings > 300
On Axis 65030 **Q2 18427** Q4 17577 Q3 14519 Q1 9418	**Q4 391** Q2 311 Q3 206 On Axis 168 Q1 22

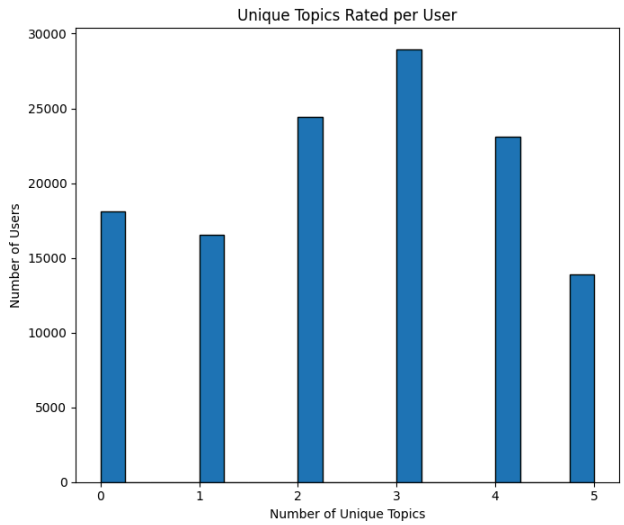
But looking at quadrants, even among highly active users (300+), the majority fall in partisan quadrants (Q2+Q4 = 702) vs on-axis (168), and the partisan lean flips. At low activity pro-Republican slightly dominates, but at high activity pro-Democrat slightly dominates (391 vs 311).

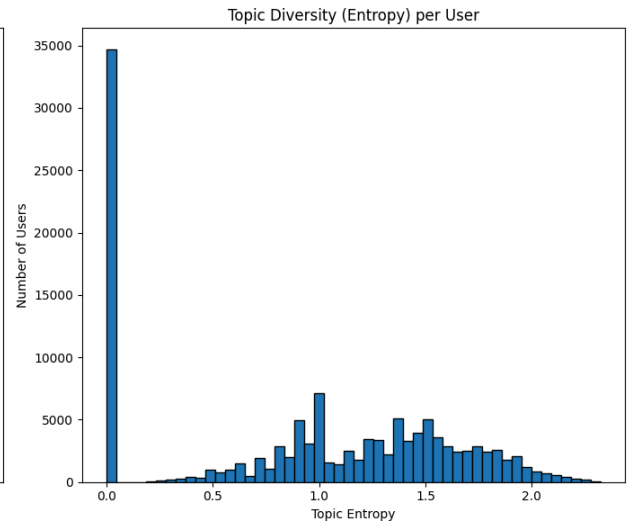
This might mean that people who are more engaged with rating on Community Notes tend to be more left-leaning overall. A possible explanation could be that Democrat-leaning users are more motivated to fact-check at high volumes, while Republican-leaning users engage more casually.

4. Do users who rate one side helpful and the other side not helpful tend to have a difference in *how frequently* they rate notes from each side?

Interests

5. Do users tend to focus on a single topic, or do they tend to mix topics?





For unique topics rated per user, the mean value was 2.5 with standard deviation of 1.56. There were 16534 users with only 1 unique topic, which is 13.23% of all users with greater than 10 ratings. For entropy per user, the spike at entropy = 0 includes both genuine single-topic raters and users whose ratings had no topic match in the political dataset.

6. What is the relationship between how frequently users contribute and what topics they focus on?

Skill

6. What is the relationship between how frequently users contribute and their skill?